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### Golf club head impact location based on 3D magnetic field readings

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#### Abstract

A 3-axis magnetometer inside a golf ball to detect the impact location on golf club heads that contain ferrous materials is disclosed herein. Key aspects of a golf shot are determined from within the golf ball itself. A magnetometer, preferably running at 85 Hz, inside a golf ball is able to measure spins of 5000 RPM. An integrated circuit comprises a gyroscope, a magnetometer, and a BLUETOOTH low energy (BTLE) radio, and at least one battery. A body is composed of an epoxy material, and the body encompasses the integrated circuit.

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## **Background/Summary**

CROSS REFERENCE TO RELATED APPLICATION (1) The Present Application claims priority to U.S. Patent Application No. 63/230,904, filed on Aug. 9, 2021, and the Present Application is a continuation-in-part application of U.S. patent application Ser. No. 17/485,164, filed on Sep. 24, 2021, which is a continuation application of U.S. patent application Ser. No. 17/162,072, filed on Jan. 29, 2021, now U.S. patent Ser. No. 11/344,785, issued on May 31, 2022, which is a continuation application of U.S. patent application Ser. No. 16/814,751, filed on Mar. 10, 2020, now U.S. patent Ser. No. 10/918,929, issued on Feb. 16, 2021, which is a continuation application of U.S. patent application Ser. No. 16/509,232, filed on Jul. 11, 2019, now U.S. patent Ser. No. 10/688,366, issued on Jun. 23, 2020, which claims priority to U.S. Provisional Patent Application No. 62/697,584, filed on Jul. 13, 2018, which is hereby incorporated by reference in its entirety.

## **STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT**

(1) Not Applicable

## **BACKGROUND OF THE INVENTION**

### **Field of the Invention**

(2) The present invention relates determining an impact location of a golf ball on a golf club during a golf swing.

### **Description of the Related Art**

(3) Most patents that have been filed looking at communicating between a ball and a device involve only trying to find the golf ball using RFID type circuitry. Most of the designs will only be successful in getting a user close to the position of the golf ball.

(4) In recent years the available technology and interest in measuring, visualizing, understanding, reviewing, and utilizing data on a golf shot has increased. Golf simulators are more accurate and “true to life” through the use of technologies such as Trackman and GC Quad. Technologies like

these are used to aid players on a driving range by providing feedback and information about a given shot. Practice ranges at PGA Tour events are full of professionals checking their performance with coaches and caddies to improve, refine, and understand their performance. Instructors and retailers use advanced golf simulators to fix a swing flaw or recommend the optimum golf club or clubs or ball for an individual. Televised golf events display launch and trajectory data for a given shot, including shot ball speed, launch angle, and spin rate. All of these measurements are, however, external to the golf ball being hit. Furthermore, these technologies are expensive.

(5) To the extent that electronics are within a golf ball, they are not capable of any measurement, but rather are used for identification purposes, as in the RFID technology used in driving ranges that track where a golf shot is collected (such as at Top Golf). Furthermore, creating a golf ball with electronics inside poses concerns of ball durability and reproducibility of the feel of a normal golf ball.

(6) In regards to the spin measurement, most spin measurement devices use Doppler technology to measure the ball as it spins, this method produces inconsistent results that have aliasing issues at times.

## BRIEF SUMMARY OF THE INVENTION

(7) The Present Invention preferably utilizes a three-axis magnetometer inside a golf ball to detect the impact location on a golf club head that contains ferrous materials. The present invention measures the way the magnetic field of the head affects each of the three-axis of the magnetometer during impact with a golf ball. The difference between the values at rest will determine an offset depending on the location of the golf ball when it is close to the golf club head. All three-axis of the magnetometer inside the golf ball are measured at rest values (away from club). The magnetic orientation of the ball is then determined. The values are re-measured to determine the magnetic vector at impact. The offset is calculated from the difference between the magnetic orientation and the vector at impact. The offset determines a signature for one of each ball impact locations: Center, Heel, Toe.

(8) One aspect of the present invention is a method for golf club head impact location based on magnetic field readings. The method includes measuring a magnetic vector for a plurality of axis of a magnetometer within a golf ball while the golf ball is at rest. The method also includes determining a magnetic orientation of the golf ball. The method also includes measuring the magnetic vector for each of a plurality of axis of a magnetometer within the golf ball at impact with a golf club head. The method also includes calculating an offset of the impact from a difference between the magnetic orientation of the golf ball and the magnetic vector at impact. The offset determines a signature for one of each of a plurality of golf ball impact locations

(9) Having briefly described the present invention, the above and further objects, features and advantages thereof will be recognized by those skilled in the pertinent art from the following detailed description of the invention when taken in conjunction with the accompanying drawings.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 is an illustration of a golfer hitting a golf ball with internal circuitry according to the present invention therein.

(2) FIG. 2 is a block diagram of the internal circuitry.

(3) FIG. 3 is a cross-sectional view of a golf ball with an internal circuitry therein.

(4) FIG. 4 is a block diagram of components of a mobile device.

(5) FIG. 5 is a circuit diagram.

(6) FIG. 5A is a circuit diagram.

(7) FIG. 5B is a circuit diagram.

- (8) FIG. 5C is a circuit diagram.
- (9) FIG. 5D is a circuit diagram.
- (10) FIG. 5E is a circuit diagram.
- (11) FIG. 5F is a circuit diagram.
- (12) FIG. 6 is a top plan view of a flexible circuit board.
- (13) FIG. 7 is a bottom plan view of a flexible circuit board.
- (14) FIG. 8 is an illustration of an electronic component.
- (15) FIG. 9 is an illustration of an electronic component within an epoxy sphere for a golf ball.
- (16) FIG. 10 is an illustration of a flexible circuit board wrapped around multiple batteries.
- (17) FIG. 10A is an illustration of a flexible circuit board wrapped around multiple batteries within an epoxy sphere for a golf ball.
- (18) FIG. 11 is an exploded partial cut-away view of a golf ball.
- (19) FIG. 12 is top perspective view of a golf ball.
- (20) FIG. 13 is a cross-sectional view of a core component of a golf ball.
- (21) FIG. 14 is a cross-sectional view of a core component and a mantle component of a golf ball.
- (22) FIG. 15 is a cross-sectional view of an inner core layer, an outer core layer, an inner mantle layer, an outer mantle layer and a cover layer of a golf ball.
- (23) FIG. 15A is a cross-sectional view of an inner core layer, an intermediate core layer, an outer core layer, a mantle layer and a cover layer of a golf ball.
- (24) FIG. 16 is a cross-sectional view of an inner core layer under a 100 kilogram load.
- (25) FIG. 17 is a cross-sectional view of a core under a 100 kilogram load.
- (26) FIG. 18 is a cross-sectional view of a core component and a mantle component of a golf ball.
- (27) FIG. 19 is a cross-sectional view of a core component, the mantle component and a cover layer of a golf ball.
- (28) FIG. 20 is an exploded partial cut-away view of a four-piece golf ball.
- (29) FIG. 21 is an exploded partial cut-away view of a three-piece golf ball.
- (30) FIG. 22 is an exploded partial cut-away view of a two-piece golf ball.
- (31) FIG. 23 is a cross-sectional view of a two-piece golf ball.
- (32) FIG. 24 is a cross-sectional view of a three-piece golf ball.
- (33) FIG. 25 is an exploded partial cut-away view of a three-piece golf ball.
- (34) FIG. 26 is a cross-sectional view of a three-piece golf ball with a dual core and a cover.
- (35) FIG. 27 is a cross-sectional view of a three-piece golf ball with a core, mantle and cover.
- (36) FIG. 28 is a cross-sectional view of a four-piece golf ball with a dual core, mantle layer and a cover.
- (37) FIG. 29 is a cross-sectional view of a four-piece golf ball with a core, dual mantle layers and a cover.
- (38) FIG. 30 is an illustration of the three axis of the magnetometer inside the golf ball at rest values.
- (39) FIG. 31 is an illustration of a measurement of the orientation of the golf ball with a center impact.
- (40) FIG. 32 is an illustration of the three axis of the magnetometer inside the golf ball for the center impact of FIG. 31.
- (41) FIG. 33 is an illustration of a measurement of the orientation of the golf ball with a toe impact.
- (42) FIG. 34 is an illustration of the three axis of the magnetometer inside the golf ball for the toe impact of FIG. 33.
- (43) FIG. 35 is an illustration of a measurement of the orientation of the golf ball with a heel impact.
- (44) FIG. 36 is an illustration of the three axis of the magnetometer inside the golf ball for the heel impact of FIG. 35.
- (45) FIG. 37 is an illustration of a calibration surface.

(46) FIG. **38** is an illustration of a ball containing the magnetometer sensor in its center is placed behind the calibration surface on an alignment orientation defining what will be the impact point on the ball.

(47) FIG. **39** is an illustration of a ball containing the magnetometer sensor in its center is placed behind the calibration surface on an alignment orientation defining what will be the impact point on the ball.

(48) FIG. **40** is a chart.

(49) FIG. **41** is an illustration of an impact detected, the magnetic field reading right before impact is used to calculate the impact location.

#### DETAILED DESCRIPTION OF THE INVENTION

(50) Use of a 3-axis magnetometer inside a golf ball and a small reference magnet to detect the impact location on a golf club face.

(51) A magnetometer is a sensor that measures the strength and direction of a magnetic field. The magnetic field measured will be a combination of both the earth's magnetic field and any magnetic field created by nearby objects. The difference between the sensor readings at rest and the readings at impact will determine an offset depending on the location of the ball when it is close to the club head.

(52) The reference magnet **74** is a small magnet **74** that is installed right behind the face **76** of the club and will be used to generate a reference magnetic field.

(53) A calibration surface is shown in FIG. **37**. The calibration surface shows X and Y coordinates with slots (circles) for a magnet for measurements.

(54) A calibration surface **72** is defined, as shown in FIG. **38**. In that surface **72**, the X and Y coordinate values and each slot (represented by P1-P8) for the magnet **74** for the measurements are shown.

(55) The ball **10** containing the magnetometer sensor **70** in its center is placed behind the calibration surface **72**, at **P0**, on an alignment orientation defining what will be the impact point on the ball **10**. The magnet **74** is removed and the measurements are taken. These readings represent the earth's magnetic field and any other magnetic interference that could be present nearby. The reference magnet **74** is then placed on each of the X, Y locations and the values are measured by the magnetometer, as shown in FIG. **39** at **P4**.

(56) FIG. **40** shows a table of calibration values.

(57) The X, Y, Z magnetic field readings are subtracted from the measurements taken without the magnet present. The values are then normalized and saved with their X, Y coordinates. These values can be interpolated for better precision.

(58) Impact location measurements.

(59) The ball **10** is aligned with the predefined impact point **78** facing the club head (face **76** of the head), as shown in FIG. **41**. The magnetic field values are collected with the ball **10** in its aligned position and away from the club that contains the reference magnet **74**.

(60) The ball **10** starts to collect magnetic field and acceleration data at a high sampling rate. When an impact is detected, the magnetic field reading right before impact is used to calculate the impact location. The X, Y, Z magnetic field readings are subtracted from the measurements taken without the magnet present. The values are then normalized and compared to the XYZ values from the calibration table, shown in FIG. **40**. The closest values define their matching X, Y coordinates.

(61) Typically, key descriptive data regarding a golf shot are captured using a device or apparatus that is external to the golf ball itself. These systems, such as Trackman or GC Quad, for example, are expensive. RFID or similar technology that is used for golf ball identification purposes does not provide any information on the club-ball impact itself or the ball's launch and trajectory information (speed, spin, angle, et cetera). It is difficult to manufacture a golf ball with electronics inside that remain functional after one or more actual golf club impacts, and replicating the feel of a normal golf ball is difficult when it contains electronics.

(62) A main objective is being able to measure key aspects of a golf shot from within the golf ball itself.

(63) Another objective is being able to extract the measurements using a simple device, such as a smartphone, rather than conducting the measurements using an expensive apparatus.

(64) The two main advantages to the consumer will be a golf ball that records spin and a golf ball that can be easily found.

(65) A magnetometer, preferably running at 85 Hz, inside a golf ball is able to measure spins of 5000 RPM. Measuring higher spin rates is also possible.

(66) The entire circuitry is preferably inside a hard plastic molded sphere.

(67) Data is transferred via BLE radio to a mobile device (in this case a phone).

(68) The circuitry inside the ball preferably activates at impact using a shock switch for power savings. At rest, after the shot, the ball keeps sending the data and going back to sleep mode every second until the user finds it using the mobile device and acknowledges it in the application.

(69) A golf ball is found using triangulation of the RSSI from the golf ball to the mobile device. The user will be instructed to move forward and to the side to generate enough space for the triangulation.

(70) Internal circuitry is embedded within the golf ball. The internal circuitry comprises at least a BLUETOOTH Low Energy radio (5th generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory. A KIONIX chip is preferred. The 5th generation BLUETOOTH Low Energy radio has a range of at least 700 meters. Triangulation is used to find a golf ball on course. The battery is preferably a 2032 coin cell. A NF52 Nordic processor is preferably utilized. A KIONIX 3-axis accelerometer is preferably utilized.

(71) As shown in FIG. 1, a golfer **100** swings a golf club **101** to hit a golf ball **10** with internal circuitry according to the present invention therein. A mobile device **120**, such as a mobile phone, receives a BLUETOOTH low energy wireless communication transmission from the golf ball **10**.

(72) FIG. 2 is a block diagram of the internal circuitry within the inner core **12a** of the golf ball **10**. The internal circuitry preferably includes a CPU **200**, a BTLE radio **201**, a memory **202**, a battery **203**, a magnetometer **204** and an accelerometer **205**.

(73) FIG. 3 is a cross-sectional view of a golf ball with an internal circuitry therein. The inner core **12a** is preferably composed of an epoxy material.

(74) FIG. 4 is a block diagram of components of a mobile device **120**. The mobile device **120** preferably comprises an accelerometer **301**, an input/output module **302**, a microphone **303**, a speaker **304**, a GPS **305**, a BLUETOOTH transceiver **306**, a WiFi transceiver **307**, a 3G/4G transceiver **308**, a RAM memory **309**, a main processor **310**, an operating system (OS) module **311**, an applications module **312**, a flash memory **313**, a SIM card **314**, a LCD display **315**, a camera **316**, a power management module **317**, a battery **318**, a magnetometer **319**, a gyroscope **320a**, LPDDR module **511**, a e-MMC module **512**, a flash module **513**, and a MCP module **514**.

(75) FIGS. 5, 5A and 5B illustrate circuit diagrams of the internal circuitry of the golf ball **10**. The internal circuitry preferably includes a CPU **200**, an antenna **211**, a first crystal oscillator **212**, a second crystal oscillator (XTAL SMD 2016, 32 MHz) **213**, an inductor (3.3 nH) **214**, a resistor **215**, a first capacitor (12 picoFaradays “pF”) **221**, a second capacitor (12 pF) **222**, a third capacitor (100 nano Faradays “nF”) **223**, a fourth capacitor (100 nF) **224**, a fifth capacitor (4.7 microFaradays “uF”) **225**, a sixth capacitor (100 nF) **226**, a seventh capacitor (12 pF) **227**, an eighth capacitor (12 pF) **228**, a ninth capacitor (100 pF) **229**, a tenth capacitor (100 pF) **230**, an eleventh capacitor (100 nF) **231**, a twelfth capacitor (NS) **232**, and a thirteenth capacitor (NS) **233**.

(76) FIG. 5C is a circuit diagram of magnetometer/accelerometer **204**, preferably a medium-G, wide bandwidth tri-axis magnetometer/tri-axis accelerometer.

(77) FIG. 5D is a circuit diagram for a gyroscope **206**, preferably a BOSCH SENSORTEC BMG250 gyroscope.

(78) FIG. 5E is a circuit diagram of a battery terminal.

(79) FIG. 5F is a circuit diagram of programming test points.

(80) FIG. 6 is a top plan view of a flexible circuit board 125.

(81) FIG. 7 is a bottom plan view of a flexible circuit board 125.

(82) FIG. 8 is an illustration of a folded flexible circuit board 125.

(83) FIG. 9 is an illustration of a folded flexible circuit board 125 within an epoxy sphere core 112a of a golf ball.

(84) FIG. 10 is an illustration of a flexible circuit board 125 wrapped around multiple batteries 130 and connected to the batteries 130 by contacts 126 and 127.

(85) FIG. 10A is an illustration of a flexible circuit board 125 wrapped around multiple batteries 130 and connected to the batteries 130 by contacts 126 and 127, and within an epoxy sphere core 112a for a golf ball.

(86) One embodiment is a golf ball 10 comprising an epoxy sphere 112a, a core layer and a cover layer. The epoxy sphere 112a comprises a body and at least one electrical component 125. The electrical component preferably comprises a plurality of stacked circuit boards and at least one battery 130 disposed within the plurality of stacked circuit boards. The body is preferably composed of an epoxy material. The body encompasses the electrical component. The core layer is disposed on the epoxy sphere. The cover layer is disposed over the core layer.

(87) The core layer preferably comprises polybutadiene material and a graphene material in an amount ranging from 0.1 to 5.0 weight percent of the outer core, wherein the outer core has a flexural modulus ranging from 80 MPa to 95 MPa.

(88) The plurality of stacked circuit boards preferably comprises an integrated circuit, a gyroscope, a magnetometer, and an antenna.

(89) The electrical component preferably has a width ranging from 5 to 20 mm, a height ranging from 5-20 mm and a length ranging from 5-20 mm.

(90) The epoxy sphere preferably has a diameter ranging from 0.4 inch to 0.9 inch, and more preferably a diameter ranging from 0.45 inch to 0.6 inch.

(91) The integrated circuit is preferably flexible and is wrapped around the at least one battery.

(92) The integrated circuit is attached to the at least one battery at three contact points.

(93) The electrical component is preferably centered within the epoxy sphere.

(94) The integrated circuit comprises a BLUETOOTH antenna, a 1 GigaHertz antenna, a microcontroller and a radiofrequency transceiver.

(95) The integrated circuit preferably comprises a plurality of capacitors and at least one inductor.

(96) The electrical component preferably detects a spin of the golf ball and transmits a signal to a mobile device.

(97) FIGS. 11, 13, 14 and 15 illustrate a five piece golf ball 10 comprising an inner core 12a, an outer core 12b, an inner mantle 14a, an outer mantle 14b, and a cover 16, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory.

(98) FIG. 15A illustrates a five piece golf ball 10 comprising an inner core 12a, an intermediate core 12c, an outer core 12b, a mantle 14, and a cover 16.

(99) FIGS. 18 and 19 illustrate a six piece golf ball 10 comprising an inner core 12a, an intermediate core 12c, an outer core 12b, an inner mantle 14a, an outer mantle 14b, and a cover 16, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory.

(100) FIG. 20 illustrates a four piece golf ball comprising a dual core, a boundary layer and a cover, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory.



(101) FIG. 21 illustrates a three piece golf ball comprising a core, a boundary layer and a cover, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory.

(102) FIGS. 22 and 23 illustrate a two piece golf ball 10 with a core 12 and a cover 16 formed of a sprayed polyurea with a thickness ranging from 0.010 inch to 0.040 inch.

(103) FIGS. 24 and 25 illustrate a three-piece golf ball 5 comprising a core 12, a mantle layer 14 and a cover 16 with dimples 18, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory.

(104) FIG. 26 illustrates a dual core three piece golf ball 35 comprising an inner core 30, and outer core 32 and a cover 34, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory h.

(105) FIG. 27 illustrates a three piece golf ball 45 comprising a core 40, a mantle layer 42 and a cover 44, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory.

(106) FIG. 28 illustrates a dual core four piece golf ball 55 comprising an inner core 51, an outer core 52, a mantle layer 54 and a cover 56, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory.

(107) FIG. 29 illustrates a four piece golf ball 65 comprising a core 60, an inner mantle 62, an outer mantle 64 and a cover 66, with an internal circuitry comprising at least a BLUETOOTH Low Energy radio (5 generation), a processor, a magnetometer, an accelerometer, and a battery. The internal circuit may also have a memory.

(108) The mantle component is preferably composed of the inner mantle layer and the outer mantle layer. The mantle component preferably has a thickness ranging from 0.05 inch to 0.15 inch, and more preferably from 0.06 inch to 0.08 inch. The outer mantle layer is preferably composed of a blend of ionomer materials. One preferred embodiment comprises SURLYN 9150 material, SURLYN 8940 material, a SURLYN AD1022 material, and a masterbatch. The SURLYN 9150 material is preferably present in an amount ranging from 20 to 45 weight percent of the cover, and more preferably 30 to 40 weight percent. The SURLYN 8945 is preferably present in an amount ranging from 15 to 35 weight percent of the cover, more preferably 20 to 30 weight percent, and most preferably 26 weight percent. The SURLYN 9945 is preferably present in an amount ranging from 30 to 50 weight percent of the cover, more preferably 35 to 45 weight percent, and most preferably 41 weight percent. The SURLYN 8940 is preferably present in an amount ranging from 5 to 15 weight percent of the cover, more preferably 7 to 12 weight percent, and most preferably 10 weight percent.

(109) SURLYN 8320, from DuPont, is a very-low modulus ethylene/methacrylic acid copolymer with partial neutralization of the acid groups with sodium ions. SURLYN 8945, also from DuPont, is a high acid ethylene/methacrylic acid copolymer with partial neutralization of the acid groups with sodium ions. SURLYN 9945, also from DuPont, is a high acid ethylene/methacrylic acid copolymer with partial neutralization of the acid groups with zinc ions. SURLYN 8940, also from DuPont, is an ethylene/methacrylic acid copolymer with partial neutralization of the acid groups with sodium ions.

(110) The inner mantle layer is preferably composed of a blend of ionomers, preferably comprising a terpolymer and at least two high acid (greater than 18 weight percent) ionomers neutralized with sodium, zinc, magnesium, or other metal ions. The material for the inner mantle layer preferably has a Shore D plaque hardness ranging preferably from 35 to 77, more preferably from 36 to 44, a

most preferably approximately 40. The thickness of the outer mantle layer preferably ranges from 0.025 inch to 0.050 inch, and is more preferably approximately 0.037 inch. The mass of an insert including the dual core and the inner mantle layer preferably ranges from 32 grams to 40 grams, more preferably from 34 to 38 grams, and is most preferably approximately 36 grams. The inner mantle layer is alternatively composed of a HPF material available from DuPont. Alternatively, the inner mantle layer **14b** is composed of a material such as disclosed in Kennedy, III et al., U.S. Pat. No. 7,361,101 for a Golf Ball And Thermoplastic Material, which is hereby incorporated by reference in its entirety.

(111) The outer mantle layer is preferably composed of a blend of ionomers, preferably comprising at least two high acid (greater than 18 weight percent) ionomers neutralized with sodium, zinc, or other metal ions. The blend of ionomers also preferably includes a masterbatch. The material of the outer mantle layer preferably has a Shore D plaque hardness ranging preferably from 55 to 75, more preferably from 65 to 71, and most preferably approximately 67. The thickness of the outer mantle layer preferably ranges from 0.025 inch to 0.040 inch, and is more preferably approximately 0.030 inch. The mass of the entire insert including the core, the inner mantle layer and the outer mantle layer preferably ranges from 38 grams to 43 grams, more preferably from 39 to 41 grams, and is most preferably approximately 41 grams.

(112) In an alternative embodiment, the inner mantle layer is preferably composed of a blend of ionomers, preferably comprising at least two high acid (greater than 18 weight percent) ionomers neutralized with sodium, zinc, or other metal ions. The blend of ionomers also preferably includes a masterbatch. In this embodiment, the material of the inner mantle layer has a Shore D plaque hardness ranging preferably from 55 to 75, more preferably from 65 to 71, and most preferably approximately 67. The thickness of the outer mantle layer preferably ranges from 0.025 inch to 0.040 inch, and is more preferably approximately 0.030 inch. Also in this embodiment, the outer mantle layer **14b** is composed of a blend of ionomers, preferably comprising a terpolymer and at least two high acid (greater than 18 weight percent) ionomers neutralized with sodium, zinc, magnesium, or other metal ions. In this embodiment, the material for the outer mantle layer **14b** preferably has a Shore D plaque hardness ranging preferably from 35 to 77, more preferably from 36 to 44, a most preferably approximately 40. The thickness of the outer mantle layer preferably ranges from 0.025 inch to 0.100 inch, and more preferably ranges from 0.070 inch to 0.090 inch.

(113) In yet another embodiment wherein the inner mantle layer is thicker than the outer mantle layer and the outer mantle layer is harder than the inner mantle layer, the inner mantle layer is composed of a blend of ionomers, preferably comprising a terpolymer and at least two high acid (greater than 18 weight percent) ionomers neutralized with sodium, zinc, magnesium, or other metal ions. In this embodiment, the material for the inner mantle layer has a Shore D plaque hardness ranging preferably from 30 to 77, more preferably from 30 to 50, and most preferably approximately 40. In this embodiment, the material for the outer mantle layer has a Shore D plaque hardness ranging preferably from 40 to 77, more preferably from 50 to 71, and most preferably approximately 67. In this embodiment, the thickness of the inner mantle layer preferably ranges from 0.030 inch to 0.090 inch, and the thickness of the outer mantle layer ranges from 0.025 inch to 0.070 inch.

(114) Preferably the inner core has a diameter ranging from 0.75 inch to 1.20 inches, more preferably from 0.85 inch to 1.05 inch, and most preferably approximately 0.95 inch. Preferably the inner core **12a** has a Shore D hardness ranging from 20 to 50, more preferably from 25 to 40, and most preferably approximately 35. Preferably the inner core is formed from a polybutadiene, zinc diacrylate, zinc oxide, zinc stearate, a peptizer and peroxide. Preferably the inner core has a mass ranging from 5 grams to 15 grams, 7 grams to 10 grams and most preferably approximately 8 grams.

(115) Preferably the outer core has a diameter ranging from 1.25 inch to 1.55 inches, more preferably from 1.40 inch to 1.5 inch, and most preferably approximately 1.5 inch. Preferably the

inner core has a Shore D surface hardness ranging from 40 to 65, more preferably from 50 to 60, and most preferably approximately 56. Preferably the inner core is formed from a polybutadiene, zinc diacrylate, zinc oxide, zinc stearate, a peptizer and peroxide. Preferably the combined inner core and outer core have a mass ranging from 25 grams to 35 grams, 30 grams to 34 grams and most preferably approximately 32 grams.

(116) Preferably the inner core has a deflection of at least 0.230 inch under a load of 220 pounds, and the core has a deflection of at least 0.080 inch under a load of 200 pounds. As shown in FIGS. **16** and **17**, a mass **50** is loaded onto an inner core and a core. As shown in FIGS. **16** and **17**, the mass is 100 kilograms, approximately 220 pounds. Under a load of 100 kilograms, the inner core preferably has a deflection from 0.230 inch to 0.300 inch. Under a load of 100 kilograms, preferably the core has a deflection of 0.08 inch to 0.150 inch. Alternatively, the load is 200 pounds (approximately 90 kilograms), and the deflection of the core **12** is at least 0.080 inch. Further, a compressive deformation from a beginning load of 10 kilograms to an ending load of 130 kilograms for the inner core ranges from 4 millimeters to 7 millimeters and more preferably from 5 millimeters to 6.5 millimeters. The dual core deflection differential allows for low spin off the tee to provide greater distance, and high spin on approach shots.

(117) In an alternative embodiment of the golf ball shown in FIG. **15A**, the golf ball **10** comprises an inner core **12a**, an intermediate core **12b**, an outer core **12b**, a mantle **14** and a cover **16**. The golf ball **10** preferably has a diameter of at least 1.68 inches, a mass ranging from 45 grams to 47 grams, a COR of at least 0.79, a deformation under a 100 kilogram loading of at least 0.07 mm.

(118) In one embodiment, the golf ball comprises a core, a mantle layer and a cover layer. The core comprises an inner core sphere, an intermediate core layer and an outer core layer. The inner core sphere comprises a polybutadiene material and has a diameter ranging from 0.875 inch to 1.4 inches. The intermediate core layer is composed of a highly neutralized ionomer and has a Shore D hardness less than 40. The outer core layer is composed of a highly neutralized ionomer and has a Shore D hardness less than 45. A thickness of the intermediate core layer is greater than a thickness of the outer core layer. The mantle layer is disposed over the core, comprises an ionomer material and has a Shore D hardness greater than 55. The cover layer is disposed over the mantle layer comprises a sprayed polyurea with a thickness ranging from 0.010 inch to 0.040 inch. The golf ball has a diameter of at least 1.68 inches. The mantle layer is harder than the outer core layer, the outer core layer is harder than the intermediate core layer, the intermediate core layer is harder than the inner core sphere, and the cover layer is softer than the mantle layer.

(119) In another embodiment, shown in FIGS. **18** and **19**, the golf ball **10** has a multi-layer core and multi-layer mantle. The golf ball includes a core, a mantle component and a cover layer. The core comprises an inner core sphere, an intermediate core layer and an outer core layer. The inner core sphere comprises a polybutadiene material and has a diameter ranging from 0.875 inch to 1.4 inches. The intermediate core layer is composed of a highly neutralized ionomer and has a Shore D hardness less than 40. The outer core layer is composed of a highly neutralized ionomer and has a Shore D hardness less than 45. A thickness of the intermediate core layer is greater than a thickness of the outer core layer **12c**. The inner mantle layer is disposed over the core, comprises an ionomer material and has a Shore D hardness greater than 55. The outer mantle layer is disposed over the inner mantle layer, comprises an ionomer material and has a Shore D hardness greater than 60. The cover layer is disposed over the mantle component, comprises a sprayed polyurea with a thickness ranging from 0.010 inch to 0.040 inch. The golf ball has a diameter of at least 1.68 inches. The outer mantle layer is harder than the inner mantle layer, the inner mantle layer is harder than the outer core layer, the outer core layer is harder than the intermediate core layer, the intermediate core layer is harder than the inner core sphere, and the cover layer is softer than the outer mantle layer.

(120) In a particularly preferred embodiment of the invention, the golf ball preferably has an aerodynamic pattern such as disclosed in Simonds et al., U.S. Pat. No. 7,419,443 for a Low Volume

Cover For A Golf Ball, which is hereby incorporated by reference in its entirety. Alternatively, the golf ball has an aerodynamic pattern such as disclosed in Simonds et al., U.S. Pat. No. 7,338,392 for An Aerodynamic Surface Geometry For A Golf Ball, which is hereby incorporated by reference in its entirety.

(121) Various aspects of the present invention golf balls have been described in terms of certain tests or measuring procedures. These are described in greater detail as follows.

(122) As used herein, "Shore D hardness" of the golf ball layers is measured generally in accordance with ASTM D-2240 type D, except the measurements may be made on the curved surface of a component of the golf ball, rather than on a plaque. If measured on the ball, the measurement will indicate that the measurement was made on the ball. In referring to a hardness of a material of a layer of the golf ball, the measurement will be made on a plaque in accordance with ASTM D-2240. Furthermore, the Shore D hardness of the cover is measured while the cover remains over the mantles and cores. When a hardness measurement is made on the golf ball, the Shore D hardness is preferably measured at a land area of the cover.

(123) As used herein, "Shore A hardness" of a cover is measured generally in accordance with ASTM D-2240 type A, except the measurements may be made on the curved surface of a component of the golf ball, rather than on a plaque. If measured on the ball, the measurement will indicate that the measurement was made on the ball. In referring to a hardness of a material of a layer of the golf ball, the measurement will be made on a plaque in accordance with ASTM D-2240. Furthermore, the Shore A hardness of the cover is measured while the cover remains over the mantles and cores. When a hardness measurement is made on the golf ball, Shore A hardness is preferably measured at a land area of the cover

(124) The resilience or coefficient of restitution (COR) of a golf ball is the constant "e," which is the ratio of the relative velocity of an elastic sphere after direct impact to that before impact. As a result, the COR ("e") can vary from 0 to 1, with 1 being equivalent to a perfectly or completely elastic collision and 0 being equivalent to a perfectly or completely inelastic collision.

(125) COR, along with additional factors such as club head speed, club head mass, ball weight, ball size and density, spin rate, angle of trajectory and surface configuration as well as environmental conditions (e.g. temperature, moisture, atmospheric pressure, wind, etc.) generally determine the distance a ball will travel when hit. Along this line, the distance a golf ball will travel under controlled environmental conditions is a function of the speed and mass of the club and size, density and resilience (COR) of the ball and other factors. The initial velocity of the club, the mass of the club and the angle of the ball's departure are essentially provided by the golfer upon striking. Since club head speed, club head mass, the angle of trajectory and environmental conditions are not determinants controllable by golf ball producers and the ball size and weight are set by the U.S.G.A., these are not factors of concern among golf ball manufacturers. The factors or determinants of interest with respect to improved distance are generally the COR and the surface configuration of the ball.

(126) The coefficient of restitution is the ratio of the outgoing velocity to the incoming velocity. In the examples of this application, the coefficient of restitution of a golf ball was measured by propelling a ball horizontally at a speed of 125 $\pm$ 5 feet per second (fps) and corrected to 125 fps against a generally vertical, hard, flat steel plate and measuring the ball's incoming and outgoing velocity electronically. Speeds were measured with a pair of ballistic screens, which provide a timing pulse when an object passes through them. The screens were separated by 36 inches and are located 25.25 inches and 61.25 inches from the rebound wall. The ball speed was measured by timing the pulses from screen 1 to screen 2 on the way into the rebound wall (as the average speed of the ball over 36 inches), and then the exit speed was timed from screen 2 to screen 1 over the same distance. The rebound wall was tilted 2 degrees from a vertical plane to allow the ball to rebound slightly downward in order to miss the edge of the cannon that fired it. The rebound wall is solid steel.

(127) As indicated above, the incoming speed should be  $125 \pm 5$  fps but corrected to 125 fps. The correlation between COR and forward or incoming speed has been studied and a correction has been made over the  $\pm 5$  fps range so that the COR is reported as if the ball had an incoming speed of exactly 125.0 fps.

(128) The measurements for deflection, compression, hardness, and the like are preferably performed on a finished golf ball as opposed to performing the measurement on each layer during manufacturing.

(129) Preferably, in a five layer golf ball comprising an inner core, an outer core, an inner mantle layer, an outer mantle layer and a cover, the hardness/compression of layers involve an inner core with the greatest deflection (lowest hardness), an outer core (combined with the inner core) with a deflection less than the inner core, an inner mantle layer with a hardness less than the hardness of the combined outer core and inner core, an outer mantle layer with the hardness layer of the golf ball, and a cover with a hardness less than the hardness of the outer mantle layer. These measurements are preferably made on a finished golf ball that has been torn down for the measurements.

(130) Preferably the inner mantle layer is thicker than the outer mantle layer or the cover layer. The dual core and dual mantle golf ball creates an optimized velocity-initial velocity ratio ( $V_i/IV$ ), and allows for spin manipulation. The dual core provides for increased core compression differential resulting in a high spin for short game shots and a low spin for driver shots. A discussion of the USGA initial velocity test is disclosed in Yagley et al., U.S. Pat. No. 6,595,872 for a Golf Ball With High Coefficient Of Restitution, which is hereby incorporated by reference in its entirety. Another example is Bartels et al., U.S. Pat. No. 6,648,775 for a Golf Ball With High Coefficient Of Restitution, which is hereby incorporated by reference in its entirety.

(131) Alternatively, the cover **16** is composed of a thermoplastic polyurethane/polyurea material. One example is disclosed in U.S. Pat. No. 7,367,903 for a Golf Ball, which is hereby incorporated by reference in its entirety. Another example is Melanson, U.S. Pat. No. 7,641,841, which is hereby incorporated by reference in its entirety. Another example is Melanson et al, U.S. Pat. No. 7,842,211, which is hereby incorporated by reference in its entirety. Another example is Matroni et al., U.S. Pat. No. 7,867,111, which is hereby incorporated by reference in its entirety. Another example is Dewanjee et al., U.S. Pat. No. 7,785,522, which is hereby incorporated by reference in its entirety.

(132) Bartels, U.S. Pat. No. 9,278,260, for a Low Compression Three-Piece Golf Ball With An Aerodynamic Drag Rise At High Speeds, is hereby incorporated by reference in its entirety.

(133) Chavan et al, U.S. Pat. No. 9,789,366, for a Graphene Core For A Golf Ball, is hereby incorporated by reference in its entirety.

(134) Chavan et al, U.S. patent Ser. No. 10/039,959, for a Graphene Core For A Golf Ball, is hereby incorporated by reference in its entirety.

(135) Chavan et al, U.S. patent Ser. No. 10/058,741, for a Carbon Nanotubes Reinforced Dual Core A Golf Ball, is hereby incorporated by reference in its entirety.

(136) Simonds et al., U.S. Pat. No. 9,707,454 for a Limited Flight Golf Ball With Embedded RFID Chip is hereby incorporated by reference in its entirety.

(137) Simonds et al., U.S. patent Ser. No. 10/252,117 for a Graphene Core Golf Ball With An Integrated Circuit is hereby incorporated by reference in its entirety.

(138) Balardeta et al., U.S. Pat. No. 8,355,869 for a Golf GPS Device is hereby incorporated by reference in its entirety.

(139) Raposo, U.S. Pat. No. 8,992,346 for a Method And System For Swing Analysis is hereby incorporated by reference in its entirety.

(140) Balardeta et al., U.S. Pat. No. 8,845,459 for a Method And System For Shot Tracking is hereby incorporated by reference in its entirety.

(141) Raposo, U.S. patent application Ser. No. 16/157,998, filed on Oct. 11, 2018, for a Smart Golf

Ball, is hereby incorporated by reference in its entirety.

(142) From the foregoing it is believed that those skilled in the pertinent art will recognize the meritorious advancement of this invention and will readily understand that while the present invention has been described in association with a preferred embodiment thereof, and other embodiments illustrated in the accompanying drawings, numerous changes, modifications and substitutions of equivalents may be made therein without departing from the spirit and scope of this invention which is intended to be unlimited by the foregoing except as may appear in the following appended claims. Therefore, the embodiments of the invention in which an exclusive property or privilege is claimed are defined in the following appended claims.

## Claims

1. A system for determining a golf club impact location with a golf ball based on three-dimensional magnetic readings, the system comprising: a golf club comprising a magnet that generates a magnetic field; a golf ball comprising a magnetometer; a mobile device comprising a mobile app; wherein the magnetometer is configured to detect the magnetic field prior to impact; wherein the magnetometer is configured to transmit a signal to the mobile app; wherein the mobile app is configured to calculate an impact location on the golf club; wherein the mobile app is configured to subtract X, Y, Z magnetic field readings from a plurality of measurements without the magnet present, normalizing the subtracted values and comparing the subtracted values to a calibration table of values to identify a match.
  2. The system according to claim 1 wherein the golf club is a driver.
  3. The system according to claim 1 wherein the magnetometer is a three-axis magnetometer.
  4. The system according to claim 1 wherein the mobile device is a tablet computer.
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